Group time against the same route flown alone

Does the straighter progress of groups remain on the same route?

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Abstract

Over a quarter of an hour a paraglider in a group progresses straighter than a solo one, but groups form on routes where any pilot may progress straight. We compare each piece of group time with the same route flown on another day, in the same season, hour and wing level, by flights that never had a contact and that no other flight came within 10 km of at the same time. On 15 539 pieces the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.017$ [$+0.011$, $+0.025$] against these controls and $+0.058$ against the solo flights of the same site, day and hour: fixing the route lowers it by 70%, by 69% for experienced pilots and 68% for beginners. The rest is small and its band excludes zero. At 100 s group time is less straight ($\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$ is -0.004), and groups spend more time climbing and less gliding, while climb rate, glide speed and glide ratio do not differ. The route accounts for most of the straighter progress; what remains goes with flying together, and whether the group causes it stays open.

1 Group or route?

Over a quarter of an hour, group time progresses straighter than solo flight. The general report (`vs-solo/general`) measures $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ of $+0.058$ against the solo flights of the same site, day and hour. That comparison leaves the route free. Groups may form on a ridge or a convergence line where any pilot progresses straight, and the straightness would then belong to the route and not to the group. Does the gap remain when group time and its control fly the same route?

This report answers with the same route flown alone, on another day. It differs from the test of the general report on three points.

- **No site condition, all the pieces.** The control may come from any site and date other than that of the piece, and every piece of group time of the paragliders that has one is used, not a sample.
- **Controls that were really alone.** A flight with no contact in the grouping can still have flown near others, so the control must also have had nobody within 10 km at the same time.
- **By wing level.** A piece is compared with controls of the same level, and the results are given for experienced and beginner pilots apart.

Section 2 says how a control is chosen and how the gap is measured, Section 3 gives the gap, Section 4 the gap by wing level and Section 5 the climbs and glides.

2 The control and the measure

Which flights are the same route flown alone, and how is the gap with group time measured?

Pieces and measure. The pieces and $H_{\mathrm{loc}}(\tau)$ are those of the general report: a *piece* is a stretch of group time of a paraglider within one segment of the track, of at least 1250 s; $\mathrm{MSD}(\tau)$ is the ensemble time-averaged mean squared displacement; $H_{\mathrm{loc}}(\tau)$ is half the local slope of $\log \mathrm{MSD}$ against $\log \tau$ over the lags within ± 0.25 decades of τ. We read it at $\tau = 100$ s, within one climb or glide, and at $\tau = 1000$ s, over one to two climb–glide cycles.

Matching. A segment k of a solo flight is a control of the piece g when all hold.

(i) **Route.** At least 80% of the points of g, one every 30 s, lie within 500 m of the track of the flight of k, which flew them in the same direction; k is the part of the track between its points nearest to the first and to the last of them, and lasts at least 1250 s.

(ii) **Date.** Other than that of g, at any site.
(iii) **Season and hour.** The middles of g and k are in months within one (circular) and in local hours within two.
(iv) **Wing level.** The same: beginners (A, B) or experienced (C, D, CCC) in the FFVL catalogue.
(v) **Solo.** The flight never had a contact of any length.
(vi) **Isolation.** At every point of k, one every 30 s, no other flight of the archive, alone or in a group, paraglider or hang glider, was within 10 km within one minute (the index of the archive holds one position per minute).

All the matches are kept. 19 786 pieces have a match under the first five conditions and 15 539 after the isolation; they have 85 524 matches, a median of 3 per piece, from 5 648 solo flights of 2 000 pilots, against 3 189 pilots on the group side.

Comparison. In analogy with the zones of the general report, the stratum is the piece. The control of a piece is the mean of the MSD of its matches, and each piece weighs the same on both sides. With n pieces and m_g the matches of the piece g,

$$\mathrm{MSD}_{\mathrm{G}}(\tau) = \frac{1}{n}\sum_{g}\mathrm{MSD}_{g}(\tau), \qquad \mathrm{MSD}_{\mathrm{ctrl}}(\tau) = \frac{1}{n}\sum_{g}\frac{1}{m_g}\sum_{k\in g}\mathrm{MSD}_{k}(\tau), \tag{1}$$

and both sides are means of many MSDs before their logarithm. The gap is $\Delta H_{\mathrm{loc}}(\tau) = H^{\mathrm{G}}_{\mathrm{loc}}(\tau) - H^{\mathrm{ctrl}}_{\mathrm{loc}}(\tau)$, positive when group time progresses straighter. The baseline compares the same pieces with the solo flights of their site-day and hour, so that the two gaps refer to the same group time; the share of the baseline gap that the route explains is

$$1 - \frac{\Delta H^{\mathrm{route}}_{\mathrm{loc}}(1000\ \mathrm{s})}{\Delta H^{\mathrm{base}}_{\mathrm{loc}}(1000\ \mathrm{s})}. \tag{2}$$

Uncertainty. Bands are 5–95% of 1000 bootstrap draws that resample the pilots and the dependency blocks of the grouping, independently, and recompute every statistic; each difference has its own band. The thin lines of the figures are the range when one year at a time is left out. The baseline and the explained share are point estimates.

What we keep. The control is the same route, on another day, in the same season, hour and wing level, flown by a solo pilot with nobody within 10 km. One gap, ΔH_{loc}, is read at 100 and 1000 s and compared with the baseline of the same pieces.

3 What remains of the gap

How much of the straighter progress of groups is left once the route is the same?

Over a quarter of an hour, group time still progresses straighter than the same route flown alone: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.017$ [$+0.011$, $+0.025$], against $+0.058$ for the same pieces against the solo flights of their site-day and hour (Table 1). Fixing the route lowers the gap by 70%, Eq. (2); its band stays away from zero. The long series, on the pieces that last long enough, gives about the same gap up to 3000 s, with bands wide enough to reach zero (Figure 1).

The gap changes sign with the lag. At 100 s it is -0.004 [-0.008, -0.002]: within one climb or glide, group time is less straight than the same route flown alone, and it is more straight from about 200 s on (Figure 1).

wing level	pieces	matches	against site-day and hour	against isolated, same route	explained
all	15 539	85 524	$+0.058$	$+0.017$ [$+0.011$, $+0.025$]	70%
experienced	10 824	64 242	$+0.051$	$+0.016$ [$+0.010$, $+0.024$]	69%
beginners	4 715	21 282	$+0.068$	$+0.022$ [$+0.006$, $+0.038$]	68%

Table 1: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ of group time against the solo flights of the same site-day and hour, and against the isolated solo flights on the same route, with the 5–95% band; the share of the first that the route explains, Eq. (2).

What we keep. The route, with the season, the hour and the wing level, explains about 70% of the long-lag gap between group and solo flight. The rest, $+0.017$ at 1000 s, is small and goes with flying together.

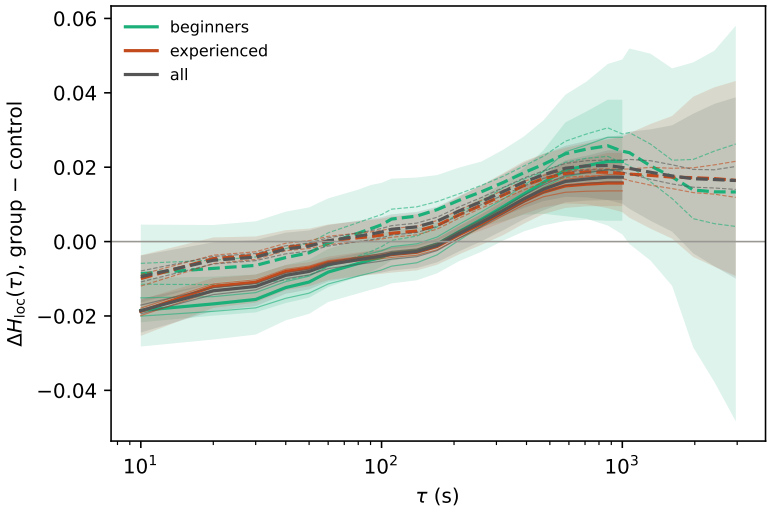


Figure 1: $\Delta H_{\mathrm{loc}}(\tau)$, group time minus the same route flown alone, for beginners, experienced pilots and both. Continuous: the lags up to 1000 s; dashed: the long series, on the pieces that last long enough. Bands 5–95%, thin lines the range when one year is left out.

4 Experienced pilots and beginners

Does the route explain the same share for both wing levels?

It does (Table 1, Figure 2). For the 10 824 pieces of experienced pilots $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from $+0.051$ to $+0.016$ [$+0.010$, $+0.024$], 69% explained. For the 4 715 pieces of beginners it falls from $+0.068$ to $+0.022$ [$+0.006$, $+0.038$], 68% explained. The beginners' baseline gap is larger and their band wider, since they have fewer pieces; the two levels differ by one point in the share explained, well within their bands.

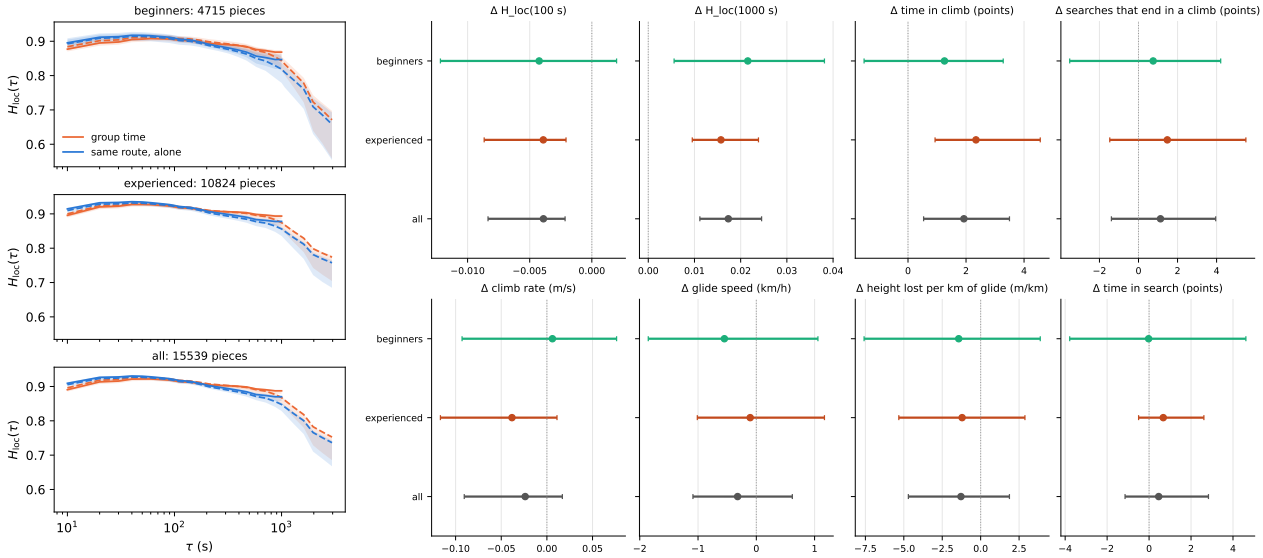


Figure 2: Left: $H_{\mathrm{loc}}(\tau)$ of group time (orange) and of the same route flown alone (blue), by wing level; dashed, the long series. Right: the difference group minus control of two H_{loc} and of the statistics of climbs and glides, with 5–95% bands, by wing level.

What we keep. Experienced pilots and beginners give the same share explained, so that the route accounts for the same part of the gap whatever the wing level.

5 Climbs and glides

Does a group climb and glide differently from the same route flown alone?

Table 2 compares the time in each phase and the statistics of climbs and glides of the pieces that have phases (loggers of 1 s) with the same statistics on their controls. Group time has more time in climb and less in glide, and its pilots start more climbs per hour; the climb rate, the glide speed, the glide ratio and

the height lost per kilometre of glide differ by less than their bands. In a group the pilot spends more time climbing, and the climb itself is not faster.

all levels	group time	same route alone	difference
time in glide (%)	57.5	59.9	-2.4 [-5.0, -1.0]
time in search (%)	13.7	13.2	$+0.5$ [-1.1, $+2.8$]
time in climb (%)	28.8	26.9	$+1.9$ [$+0.5$, $+3.5$]
searches ending in a climb (%)	49.6	48.5	$+1.1$ [-1.4, $+4.0$]
climbs per hour	7.46	6.99	$+0.48$ [$+0.20$, $+0.94$]
climb rate (m/s)	1.12	1.14	-0.02 [-0.09, $+0.02$]
glide speed (km/h)	28.6	28.9	-0.3 [-1.1, $+0.6$]
glide ratio	16.2	15.9	$+0.3$ [-0.5, $+1.2$]
height lost per km of glide (m/km)	61.6	62.8	-1.3 [-4.7, $+1.9$]

Table 2: Climbs and glides of group time and of the same route flown alone, both wing levels together; the difference group minus control with its 5–95% band. Times are shares of the time flown, in %, and their differences are in percentage points.

The right of Figure 2 gives the same differences for each level: the bands of the beginners are wider and contain zero for every statistic of the phases.

What we keep. On the same route a group spends more time climbing and less gliding; the speed, the rate and the glide ratio do not differ.

6 Conclusions and limits

What can be concluded, and what cannot?

What we keep.

- The route, with the season, the hour and the wing level, explains 70% of the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ between group time and solo flight; the same for experienced pilots and beginners.
- What remains, $+0.017$ [$+0.011$, $+0.025$], is small, with a band away from zero; at 100 s group time is less straight than the control.
- On the same route a group spends more time climbing and less gliding; the climb rate and the glide speed do not differ.

Limits.

- The days differ: season and hour are matched, the weather of the day is not.
- The controls flew with nobody within 10 km. They may differ from the pilots who fly in groups, and this test has no selection contrast to measure it.
- Only the 15 539 pieces whose route was flown alone on another day enter; their routes are frequented ones.
- The bands measure the sampling of pilots and blocks. The radius and the cover of the route were varied only in the general report, on a sample, and the isolation distance not at all.

Fixing the route removes most of the straighter progress of groups; whether what remains is caused by the group stays open.